

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Regular Examination, 2022
URP 2115
(Urban Planning Techniques)

Full Marks: 210

Time: 3 hrs

- N.B.**
- i) Figures in the right margin indicate full marks.
 - ii) Answer any **three** questions from each section in separate script.
 - iii) Assume data where necessary.

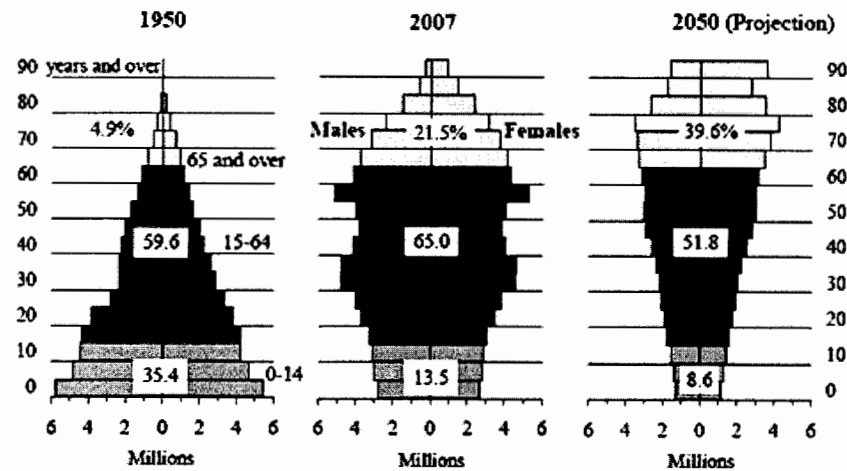
Section – A

1. (a) Write about the concept of “Urban Development Plan” in controlling unplanned urbanization of Bangladesh. (10)
(b) Briefly discuss the issues raised by the stakeholder groups of Detailed Area Plan. (10)
(c) “Detailed Area Plan is currently practiced to overcome some problems of Master Plan”. Make a summary on the feature of Detailed Area Plan. (15)
2. (a) Which technique between Guided Land Development and Land Readjustment Plan will be used for developing an urban fringe area in Bangladesh? Why? (15)
(b) “Area Improvement Plan is feasible in already developed areas of a town”- do you agree? Discuss the methods of development used in this technique. (10)
(c) Make a comparative analysis between the following: (10)
 - i. Action area plan and Subject area plan
 - ii. Land Pooling and Land Ceiling
3. (a) “Master Plan is necessary to control the development in a systematic way”- explain with proper example. (15)
(b) Write down the purposes of Detailed Area Plan. (10)
(c) “Structure Plan provides more flexible and open-ended plans”- explain this statement with appropriate example. (10)
4. (a) Write down a critical discussion on DMDP Structure Plan (2016-2035). (15)
(b) “Land ceiling is a significant tool for controlling urban development”- do you agree? Explain. (10)
(c) Specify the implementation challenges of Land Banking Technique in Bangladesh. (10)

Section – B

5. (a) What are the alternative population extrapolation curves? Explain each with respective equation and figure. (15)
(b) Define population Estimate, Projection and Forecast. “Population projections are often mislabeled as forecast” – explain. (10)
(c) Below is the population figure of Japan in 1990, 2007 and 2050. What is your understanding about the population distribution of Japan? (10)

Changes in the Population Pyramid



Source: Statistics Bureau, MIC; Ministry of Health, Labour and Welfare.

6. (a) Between Linear and Geometric curves which would give the best possible fit to explain the population growth trend of Bangladesh? Explain. (10)
- (b) What is Locational Quotient (LQ) approach? What are the assumptions of LQ approach? Using LQ approach, find out the basic and non-basic sectors of the following economic sectors of the following: (15)

SIC code	Industry	Local	National
A	Agricultural	5240	70,96,533
C	Construction	1378	4642735
D	Food production	1687	16,014
I	Hotels	124	1,33,4210
H	Government	4060	9,32,6667
Total		12,489	22,41,6159

- (c) Given the following: (10)
- Projected growth rate for industry i in a region from year t to t' = 12.5%
- Local employment in industry in year t = 2500
- Projected local employment shift for industry i for the period t to t' = 20%
- Calculate projected local employment in industry i for year t' using constant share approach and shift share approach.
7. (a) "Demolition vs. Renewal"- which do you think is the correct choice and why? (10)
- (b) Provide a brief overview on the Build-out Analysis methods. (15)
- (c) Show the levels of intervention in Urban Renewal with the help of a diagram. (10)
8. (a) Explain suitability analysis with the help of a proper example demonstrating the calculations that take place. (15)
- (b) Depict the level of intervention in Urban Renewal with the help of a diagram. (10)
- (c) Explain the Build Out Analysis methods with the help of a proper example. (10)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Regular Examination, 2022
CE 2107
(Construction Technology & Materials)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

- 1 (a) Define concrete. Classify concrete from different manners. (06)
- (b) Define segregation. What are the causes and remedial measures of segregation? (10)
- (c) What are the factors which affect the workability of concrete? Describe briefly. (15)
- (d) What is slump? Show the different types of slumps with neat sketch. (04)
- 2 (a) What is cement? What are the functions of various ingredients of cement – describe briefly. (15)
- (b) Explain the concrete temperature vs time curve and describe each of the stages. (12)
- (c) Write short notes on Hydration of cement and curing of concrete. (08)
- 3 (a) What is gradation of aggregate? Write down the basic differences between coarse aggregate and fine aggregate. (08)
- (b) Classify brick according to their strength. Discuss the factors to be observed during the construction of brick. (15)
- (c) What are the methods used for burning of brick? Which method is used in Bangladesh? Why? (12)
- 4 (a) Define the following terms- (10)
- i. Sphericity
 - ii. Fineness modulus
 - iii. Grading
 - iv. Absorption
 - v. Flaky and elongated particle
- (b) There are two different varieties of sand available in a construction site. Sieve analysis data of sand type "A" is given below. The Fineness modulus (FM) of sand type "B" is 2.60. Calculate the amount of each variety of sand required to prepare a mixture of 10 kg sand whose combined FM will be 2.90. (17)

Sieve No.	Weight retained (gm)
#4	4
#8	15
#16	170
#30	150
#40	91
#50	31
#100	25
#200	10
Pan	4

- (c) What are the functions of water in concrete? And what are the qualities of water? (08)

Section – B

- 5 (a) What is foundation? What are the causes of foundation failure? (08)
- (b) Discuss various types of deep foundation with neat sketch. (12)
- (c) Write short notes on- (15)
- i. Skin friction
 - ii. Raft foundation
 - iii. Bearing capacity
 - iv. Isolated footing
 - v. End bearing
- 6 (a) What are the functions of lintel in masonry building? (06)
- (b) Draw a typical section of reinforced concrete lintel with Chajja. (05)
- (c) Define voussoirs, key stone, skew-back of the arch. Classify arches according to their shape and briefly discuss them. (14)
- (d) Determine the safe bearing capacity of pile if a drop hammer weighing 1300 kg and falling through a height of 300 mm results in an average penetration of 0.45 cm in the last few blows. The weight of the pile with all stationary parts resulting on its head may be taken as 1000 kg. (10)
- 7 (a) What is cavity wall? What are the advantages of cavity wall? (06)
- (b) Write down the position of cavity wall and discuss about constructional details. (10)
- (c) What is stair? Discuss briefly the requirements of a good stair. (10)
- (d) Write a short note on- (09)
- i. Double hung windows
 - ii. Semi-circular arch
 - iii. Bearing joint
- 8 (a) What are the main differences between floor and roof? Write down the factors to be considered for the selection of suitable floor type. (11)
- (b) Define- (06)
- i. Soffit
 - ii. Scotia
 - iii. Nosing of stair
- (c) Draw various types of stairs (plan and elevation view) with neat sketch (10)
- (d) What are the objectives of plastering? Briefly explain various types of defects found in plastering. (08)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Regular Examination, 2022
URP 2113
(Site and Area Planning)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section A

1. (a) "Site planners designate the uses of land, arrange structure and space" – (10)
narrate the scope of site planning.
- (b) Explain the following terms: (10)
 - (i) Non-contiguous, stand-alone site
 - (ii) Sites within a larger, contiguous property.
- (c) Explain the four basic models of site planning with proper diagram. (15)
2. (a) SMART Growth can be ensured by the process of site selection. Explain this (10)
statement with necessary example.
- (b) "The Sun's specific position is the combination of two measurements"- briefly (10)
explain with necessary diagram.
- (c) Narrate the significance of land use control in urban planning. Explain the (15)
following land use control models with necessary diagrams.
 - (i) Burgess zone model
 - (ii) Hoyt sector model.
3. (a) How will you follow the guiding principles of recreation, parks and open (15)
space (RPOS) for developing a park in Khulna City?
- (b) Calculate the super elevation required for a road of 7m wide on a curve of (10)
250m radius for permissible speed of 80kmph. Let the coefficient of friction
be 0.15.
- (c) Explain the following terms with proper diagrams: (10)
 - i) Sag vertical curves
 - ii) Crest vertical curves.
4. (a) Write down in brief about the building orientation, room arrangement and (15)
shading devices needed in a residential building in warm humid zone with
proper diagram.
- (b) What is wind rose diagram? Prepare a wind rose diagram following the data (10)
in the below table:

Direction	Number of measurements
000° - 029°	6
030° - 059°	15
060° - 119°	9
120° - 149°	3

- (c) Briefly discuss about the aesthetic factors of site inventory and analysis. (10)

Section B

5. You are an Urban Planner working on a residential neighborhood planning and design project. You are instructed to create 3, 4, and 5 katha plots with a design density of 300 people per acre. Besides, you are said to follow the space standards for neighborhood layout design. The neighborhood area is 100 acres. Read the aforementioned statement and answer the following questions:
- (a) What do you understand by neighborhood planning? (05)
 - (b) According to aforementioned statement, how many primary and secondary schools are required to serve the population as per international standard? (10)
 - (c) What is the total service area of required number of primary schools, if a primary school provides service within 500m from its central point? (10)
 - (d) What is the length and width of a 3, 4 and 5 Katha plots with the golden rectangle ratio of 1:1.5. (10)
6. (a) Make a list of different types of available technologies for subdivision planning along with their applications. (10)
- (b) Write down any ten principles of neighborhood planning. (10)
- (c) Neighborhood should be planned keeping the priority of the children in mind. Write down some suggestions for child-friendly neighborhood development as an Urban Planner. (15)
7. (a) What are the regulatory bases of Private Residential Area Land Development Rules 2012? (10)
- (b) Briefly explain the model of site planning process. (10)
- (c) Write down the steps of preparing a Shadow Diagram of a site. (15)
8. (a) Write down the different patterns of subdivision plan. Briefly explain the curvilinear pattern with its positive and negative aspects. (10)
- (b) Which pattern of subdivision plan is more suitable than others in case of Bangladesh especially in the flat region as per your understanding and why? State your rationalities. (10)
- (c) Illustrate five comparative critical examples (good design and bad design) of subdivision planning and neighborhood design. (15)

Khulna University of Engineering & Technology
Department of Urban & Regional Planning
BURP 2nd Year 1st Term Examination, 2021.

URP 2191

(Statistics for Planners-I)

Full Marks: 210

Time: 3.0hrs

- N.B.** i) Figures in the right margin indicate full marks.
ii) Answer any **three** questions from each section in separate script.
iii) Assume data where necessary.

Section – A

1. (a) Explain “statistics in plural sense”. Mention the major characteristics of statistics (15) with examples.
(b) What is statistical inquiry? What are the methods of field investigation for (10) collecting primary data?
(c) Mention the key consideration of framing a questionnaire for primary data (10) collection.
2. (a) Draw a balanced and good table (assume data). Explain the major considerations (15) for a good table by listing the major parts.
(b) The sale proceeds of 50 shops are given below in Taka. Prepare frequency table (12) by taking suitable class-interval and represent the table by frequency polygon.
82,40,56,75,46,87,43,62,70,64,58,77,52,63,55,64,67,37,66,73,34,72,61,85,58,62,74,55,67,52,54,58,64,65,48,69,54,59,47,61,66,66,72,57,62,60,64,72,55,63,67.
(c) Following table shows per capita expenditures on different items of consumption (08) in two years. Compare the two years’ expenditure with suitable graph.

Item	Per capita expenditure in tk.	
	1978	1977
Food	189	164
Clothing	17	12
Housing	43	26
Miscellaneous	34	28

3. (a) The following table shows distribution of 200 shops according to sale proceeds (12) per week. Calculate quartiles and quartile deviation.

Weekly sale in Taka	No. of shops
Upto 500	15
501-1000	31
1001-1500	43
1501-2000	15
2001-2500	22
2501-3000	13
3001-3500	3
3501-4000	7
4001-4500	2
4501-5000	7
5000 and above	40

- (b) Using the following data, draw Box-Whiskers plot. (16)
5,7,2,1,3,4,8,8,6,15,23,28,32,45
Also calculate: (i). first quartile, (ii) third quartile, (iii) Interquartile range, (iv) upper and lower limits, and (v) outliers.
- (c) Graphically explain the concept of skewness and kurtosis. (07)
4. (a) From the following table: (35)

Class interval	5-4	10-14	15-19	20-24	25-29	30-34
No. of item	7	8	9	5	10	4

- (i) Draw frequency polygon in graph paper
- (ii) Draw ogive curve
- (iii) Calculate Mean, Median and Mode
- (iv) Also locate mean, median and mode in the above frequency polygon.

Section – B

5. (a) Define with example (any four) (16)
 (i) Sample space; (ii) Even; (iii) Dependent and independent even; (iv) Mutually exclusive event; (v) Conditional probability.
- (b) Four components are inspected and three event A, B, C are defined as follows: (19)
 A = all four components are either good or bad,
 B = At least one components are found to be bad,
 C = Exactly two components are good.
- (i) How many elementary event exist in the sample space?
 - (ii) Define above events A, B, C as set in algebraic way.
 - (iii) Interpret the following events: $A \cap B$, $B \cap C$, and $A \cup B \cup C$ along with its set components.
6. (a) Write short notes on the followings: (15)
 (i) Probabilistic reasoning Vs Statistical reasoning.
 (ii) Descriptive statistics Vs Inferential statistics.
- (b) Why cannot we compare the ratios of measurements with interval level data? – (10)
 Explain with examples.
- (c) A sample of employees of a planning firm is surveyed and found the following: (10)
- | Blood group | Males | Females |
|-------------|-------|---------|
| O | 20 | 20 |
| A | 17 | 18 |
| B | 8 | 7 |
| AB | 5 | 5 |
- (i) What is the probability that the first person is selected is either blood type) or blood type A?
 - (ii) What is the probability that the first person is selected is a male and blood type O.
7. (a) What is "Tree Diagram"? What is it used for? - Explain. (10)
- (b) Suppose, a rare disease infects one out of every 1000 people in a population. (15)
 Suppose, there is a good test for this disease: if a person has the disease the test comes back positive 95% of the time. One the other hand, the test also produces some false positives: 2% of the uninfected people are also test positive. What are his chances of having this disease if someone just tested positive?
- (c) How many ODD five digit numbers divisible by 5, can be male from all of the (10)
 digits, if: (a) Repetition is allowed, (b) Repetition is not allowed?
8. (a) Mention the characteristics of Normal Probability distribution and Standard (10)
 Normal Distribution.
- (b) In Tokyo City, the mean height of 10,000 building is 100m and standard (15)
 deviation 20m. If it follows a normal distribution, find how many buildings' height is:
- (i) between 75m to 125m
 - (ii) greater than 200m
 - (iii) less than 50m [indicate the probability in figure]
- (c) Scores for a civil service exam are normally distributed with a mean of 70 and a (10)
 standard deviation of 7.5. To be eligible for civil service employment, you must score in the top 7% of what is the lowest score you can earn and still be eligible for employment? [Indicate the probability in figure].

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Regular Examination, 2022
HUM 2117
(Sociology for Planners)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section A

1. (a) What do you understand by Sociology? (05)
(b) Differentiate between society and community with example. (15)
(c) What is society? What are the types of society? Explain different types of society. (15)
2. (a) "Religion is a fundamental feature of all societies"- Explain. (10)
(b) "Why are human societies stratified? (10)
(c) Why do people behave as they do? What determines human behavior? (15)
3. (a) Differentiate between Ethnocentrism and Xenocentrism. (10)
(b) State the importance of culture in moulding personality of an individual. (10)
(c) Discuss different carriers of culture with necessary example. (15)
4. (a) What is social pathology? (05)
(b) What kinds of reward are especially important in deterring deviance? (10)
(c) Explain juvenile delinquency in the light of Travis Hirschi's control theory. (20)

Section B

5. (a) What do you understand by migration? Discuss in brief the major causes of migration. (10)
(b) Is migration a social problem? Explain it with example. (10)
(c) What are the aftermaths of rural urban-migration in Bangladesh? (15)
6. (a) "Urbanism as a way of life"- Explain this view of Louis Wirth. (15)
(b) What are pattern of urbanization in Bangladesh? Find out the major urban social problem and explain any two of them. (10)
(c) What is shanty town? Why is shanty town growing rapidly in 3rd world country? (10)
7. (a) Define crime and juvenile delinquency? Differentiate between white collar crime and victimless crime. (10)
(b) What are the major causes of juvenile delinquency in Bangladesh? (10)
(c) What are the categories of juvenile delinquency? How does family environment play important role with respect to juvenile delinquency? (15)
8. (a) What do you understand by urban ecology? (10)
(b) How does over population affect urban ecology? (10)
(c) Explain the changing social structure of Bangladesh in the post-independent period. (15)